

ABDALRHMAN BADAWI

AI Engineering Student | Machine Learning & Computer Vision
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PROFESSIONAL SUMMARY

AI Engineering student at Mansoura University with hands-on experience in machine learning, deep learning, and computer vision. Skilled in building end-to-end ML pipelines using Python, Scikit-learn, and PyTorch. Proven ability to deliver real-world projects through internships and academic work, with a strong foundation in statistical modeling and data analysis.

EDUCATION

B.Sc. in Engineering — AI Engineering — Mansoura University Nov 2022 — Expected Jun 2027

GPA: 3.47 / 4.00

Relevant Coursework: Deep Learning, Computer Vision, Big Data Science, Pattern Recognition, Statistical Modeling, Algorithms & Data Structures, Relational Databases, Data Visualization, Linear Algebra, Probability & Statistics

EXPERIENCE

ML & AI Intern — Digital Egypt Pioneers Initiative (DEPI) Mar 2025 — Aug 2025

- Built and evaluated machine learning pipelines using Python, Pandas, and Scikit-learn for regression and classification tasks.
- Performed data cleaning, feature engineering, and exploratory data analysis (EDA) on structured datasets, improving model accuracy.
- Applied statistical modeling techniques and evaluated models using RMSE, R^2 , MSE, and accuracy metrics.
- Developed a comprehensive ML capstone project with a team, delivering an end-to-end analytics solution.
- Communicated analytical findings to mentors and peers through data visualizations and presentations.

PROJECTS

Automated License Plate Recognition (ALPR) — Graduation Project Oct 2024 — Present | [GitHub](#)

- Developed a Computer Vision module using deep learning and OpenCV for vehicle license plate detection and text extraction.
- Applied image preprocessing techniques (resizing, cropping, grayscale conversion, noise reduction) to improve detection reliability.
- Implemented object detection and OCR to accurately read license plate characters from live video streams.
- Integrated the vision module into a broader traffic management system in collaboration with a cross-functional team.

Heart Disease Prediction App — ML Web Application Jan 2025 | [GitHub](#)

- Built a Streamlit web application to predict heart disease risk using patient health data and clinical indicators.
- Trained and compared multiple Scikit-learn classification models, selecting the best-performing one for deployment.
- Designed an interactive UI enabling real-time predictions based on inputs such as age, cholesterol, and blood pressure.

Plant Disease Classification — Deep Learning (CNN) May 2025

- Built a CNN image classification model using PyTorch and transfer learning (ResNet-18) for plant disease detection.
- Applied data augmentation and preprocessing pipelines to improve generalization on the New Plant Diseases dataset.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix visualizations.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C

AI / ML: Supervised & Unsupervised Learning, Deep Learning, Feature Engineering, EDA, Statistical Modeling, Computer Vision

ML / DL Frameworks: Scikit-learn, PyTorch, Keras, OpenCV, TensorFlow, YOLO, Hugging Face

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Developer Tools: Jupyter Notebook, VS Code, Git / GitHub, Streamlit

Languages: Arabic (Native), English (Professional)

Soft Skills: Teamwork, Problem-Solving, Analytical Thinking, Communication

CERTIFICATIONS

- [Supervised Machine Learning: Regression and Classification](#) — DeepLearning.AI (Apr 2025)
- [Foundations: Data, Data, Everywhere](#) — Google (May 2024)
- [Python for Data Science, AI & Development](#) — IBM (Jan 2024)
- [Linear Algebra for Machine Learning and Data Science](#) — DeepLearning.AI (Nov 2023)